

Transforming curiosity through AR: An empirical study of flow-based environmental health education for elementary students

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ABSTRACT

Environmental health education faces the challenge of connecting students with the often-overlooked organisms coexisting in their everyday environments. This study developed and validated an augmented reality (AR)-enhanced mosquito vector control curriculum grounded in integrated curiosity-flow theory. The curriculum employed a two-stage design: Session 1 created knowledge gaps to induce deprivation-type (D-type) curiosity, while Session 2 provided AR exploration tools to facilitate transformation into interest-type (I-type) curiosity and foster flow experience. Using structural equation modeling with 423 fifth-grade students, we examined relationships among mosquito-related anxiety, curiosity types, flow experience, and continuance intention. Results demonstrated that anxiety significantly predicted both curiosity types (I-type: $\beta = .62$; D-type: $\beta = .54$), which in turn predicted flow experience ($R^2 = .52$). Flow significantly influenced continuance intention ($\beta = .53$, $R^2 = .28$), with both curiosity pathways showing significant mediation effects. Findings illuminate how AR technologies provide unique affordances, real-time feedback, autonomous navigation, and achievement visualization, that support theoretical mechanisms in environmental education. This research contributes an empirically validated framework integrating curiosity and flow theories for educational technology design.

1. Introduction

Environmental health education faces a critical challenge: How can educators transform abstract ecological threats into compelling learning experiences that sustain student engagement? While environmental literacy has been recognized as a crucial 21st-century competency [1], traditional didactic approaches often fail to generate the intrinsic motivation necessary for lasting behavioral change. This challenge is particularly acute in elementary education, where young students' limited life experiences make it difficult to recognize the rich biodiversity that exists within their immediate surroundings and understand how their everyday environment interconnects with diverse ecosystems.

Digital game-based learning environments offer a promising solution by leveraging psychological mechanisms that drive sustained engagement. The sustained engagement necessary for such discovery-based learning can be understood through psychological mechanisms that drive intrinsic motivation. Cognitive curiosity theory [2,3] provides a theoretical lens for understanding how learners transition from anxiety-driven knowledge seeking (deprivation-type curiosity, or D-type) to pleasure-based exploration (interest-type curiosity, or I-type).

This dual-process framework suggests that effective digital learning environments should strategically induce initial knowledge gaps to trigger D-type curiosity, then provide structured exploration tools to facilitate transformation into I-type curiosity. When coupled with flow theory's principles of challenge-skill balance [4], this approach promises to create optimal learning conditions that promote both immediate engagement and long-term persistence.

Despite growing interest in gamified environmental education, research has rarely examined how digital technologies can be specifically designed to operationalize these theoretical mechanisms. Previous studies have documented the motivational benefits of gamification [5] and the learning advantages of curiosity-driven instruction [6], yet few have investigated the precise technological affordances that enable curiosity type transformation and flow experience in real-world educational settings.

RQ. Does an AR-enhanced environmental education curriculum designed to operationalize integrated curiosity-flow theory successfully trigger the hypothesized sequential relationships among anxiety, curiosity transformation, flow experience, and continuance intention?

To bridge this gap by developing and evaluating an AR-enhanced

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dengue vector education program grounded in integrated curiosity-flow theory. Using structural equation modeling with 423 fifth-grade students, we examine whether theoretically-informed design features including IoT-enabled physical props and AR exploration tools successfully trigger the hypothesized psychological processes and ultimately enhance continuance intention. Our findings contribute to both educational technology design principles and theoretical understanding of how digital affordances support motivational processes in environmental education.

2. Literature review

2.1. Cognitive curiosity theory in digital learning environments

Cognitive curiosity theory distinguishes between two fundamental types of epistemic motivation: deprivation-type (D-type) and interest-type (I-type) curiosity [2,3]. D-type curiosity arises from perceived knowledge gaps and is characterized by feelings of tension that drive information-seeking behavior to reduce cognitive dissonance. In contrast, I-type curiosity stems from intrinsic interest and is associated with pleasurable exploration and discovery. This dual-process framework has important implications for educational technology design, as it suggests that optimal learning experiences should strategically leverage both mechanisms.

While the theoretical distinction between D-type and I-type curiosity is conceptually compelling, a critical question remains: can instructional interventions deliberately induce these curiosity states to enhance learning outcomes? Recent empirical studies provide strong evidence for the feasibility of curiosity-based pedagogy, particularly in triggering the initial D-type response that serves as a gateway to deeper engagement. A large-scale randomized controlled trial demonstrated that deliberately creating "information gaps" significantly enhanced students' curiosity, knowledge retention, and science test scores, with effects persisting into middle school [6]. Critically, the study revealed that state curiosity, momentary curiosity induced by specific situations more strongly predicted classroom learning than stable trait curiosity. This distinction is consequential: it suggests that D-type curiosity can be strategically triggered through environmental design rather than relying on students' pre-existing curiosity dispositions. For digital learning environments, this finding validates a design approach where controlled information gaps initially induce D-type tension, creating the motivational foundation for subsequent I-type exploration when learners are provided with discovery tools and autonomy.

Empirical support for the pedagogical application of curiosity theory in technology-enhanced environments has grown substantially. Murayama, et al. [7] examined how information gap cues, presented as incomplete diagrams with masked data points embedded within an online science inquiry platform, generated significantly higher D-type curiosity scores and deeper processing of ecological content compared to complete-information control conditions. Critically, the shift from D-type to I-type curiosity was moderated by the availability of autonomous exploration tools: students who could select their own investigative pathways following the information gap prompt showed stronger curiosity type transformation and reported greater intrinsic motivation. This design pattern of deliberate incompleteness followed by structured autonomy directly informs the two-session curriculum architecture adopted in the present study, providing an empirical precedent for the proposed anxiety-to-curiosity induction sequence within a digital learning context.

Once D-type curiosity is triggered and learners transition into I-type exploration, curiosity enhances memory encoding and facilitates conceptual integration, transforming fragmented information into coherent knowledge structures through a self-reinforcing progress loop: as learners perceive measurable advancement during autonomous exploration, their intrinsic motivation strengthens, reinforcing continued inquiry and cultivating scientific reasoning and metacognitive awareness

[8].

However, orchestrating this trajectory in elementary STEM education presents significant design challenges. Digital learning environments offer unique affordances by enabling dynamic adaptation through real-time feedback, personalized scaffolding, and embedded assessment. AR and IoT technologies can provide contextual hints that adjust to individual exploration patterns, maintaining optimal challenge levels while preserving autonomy. Such adaptive support addresses the delicate balance between guidance and independence, offering structure during cognitive overload yet withdrawing scaffolds as competence emerges.

2.2. Flow theory in technology-enhanced learning

Flow theory describes optimal experience characterized by complete absorption in an activity, occurring when perceived challenges match perceived skills while clear goals and immediate feedback are present. In digital learning contexts, flow theory has gained substantial research support for its potential to enhance engagement, learning outcomes, and system adoption in both formal and informal educational settings. Crucially, Csikszentmihalyi's framework identifies three preconditions for flow emergence: (1) clear proximal goals that provide direction, (2) immediate and unambiguous feedback that enables performance monitoring, and (3) a dynamic balance between perceived challenges and skills that prevents both anxiety and boredom [9].

Digital learning platforms are uniquely positioned to operationalize these flow preconditions through specific affordances: gamification elements translate abstract objectives into clear progressive goals [10,11]; adaptive sequencing dynamically recalibrates the challenge-skill equilibrium [12,13]; and real-time feedback supports continuous performance monitoring while preserving learner autonomy [12,13]. Certain technology modalities combine these affordances particularly powerfully within authentic learning contexts.

2.3. Location-based augmented reality

Among these technologies, certain modalities stand out for their ability to activate multiple flow preconditions simultaneously within authentic, real-world learning contexts. One particularly promising example is location-based augmented reality (LBAR). Location-based augmented reality (LBAR) represents a particularly compelling instantiation of flow-supporting design, as it simultaneously addresses multiple flow preconditions within authentic learning contexts. By enabling superimposition of digital content onto physical environments where learners naturally explore, LBAR creates inherently meaningful goals (exploring real spaces), provides immediate spatial feedback (digital overlays responding to physical movement), and scaffolds challenge-skill balance through context-sensitive information delivery. Unlike screen-based platforms, LBAR's spatial computing enables embodied interaction where physical navigation becomes integral to the learning experience.

A systematic review reveals positive acceptance and evident enthusiasm for LBAR's learning applications, though successful implementation requires careful integration with established pedagogical practices [14]. The technology's unique combination of physical exploration and digital augmentation creates conditions particularly conducive to flow: learners maintain awareness of their authentic environment while receiving contextualized information that sustains engagement without cognitive overload. Empirical research across diverse digital learning platforms including LBAR and other technology-enhanced environments confirms that successfully induced flow states yield multiple interconnected benefits: enhanced learning outcomes and technological self-efficacy [12,15], increased engagement and persistence through positive affect, and sustained system usage through perceived usefulness pathways [16].

However, measurement heterogeneity presents challenges, as most

studies rely on self-report questionnaires, limiting cross-study comparisons [5]. A gap exists between design intentions and empirical verification: many studies document flow phenomena without explicitly designing or testing activities that reliably induce flow [5]. These limitations highlight the need for research that intentionally designs and validates flow-inducing mechanisms in educational technology.

A small but growing body of research has moved beyond documenting flow as an outcome to explicitly designing and testing technological environments intended to trigger and sustain it. Huang, et al. [17] developed a gamified AR history tour for middle school students in which task difficulty was dynamically calibrated based on real-time performance data, directly operationalizing Csikszentmihalyi's core precondition for flow experience [9]. Using experience sampling methodology, the study confirmed that students who engaged with the adaptive AR environment reported significantly higher flow scores and longer voluntary engagement times than those using a non-adaptive version of the same tour. Importantly, the study identified real-time feedback specificity as the single strongest predictor of flow onset: students who received item-level, locationally-anchored feedback that indicated exactly which historical artifact they had correctly identified and why achieved flow significantly faster than those receiving only aggregate performance summaries. This finding has direct relevance for the present study's design logic, as the AR trigger marker system similarly provides immediate, location-specific ecological information rather than deferred or generalized feedback, a feature the current empirical results confirm as central to the curiosity-to-flow transition.

2.4. Learning anxiety as a trigger of epistemic curiosity

While curiosity and flow theories explain how engagement is sustained, a critical question remains: what initially triggers the curiosity that sets this motivational cascade in motion? In environmental education contexts, particularly when introducing novel ecological concepts to elementary students, learning anxiety the psychological tension arising from awareness of one's knowledge gaps serves as a potent catalyst for curiosity induction. Learning anxiety, distinct from generalized or trait anxiety, refers to the situational cognitive and emotional discomfort experienced when learners encounter unfamiliar academic content and become aware of the discrepancy between their current and desired knowledge states [18]. In environmental education, this may manifest when students cannot identify common schoolyard insects, discover unexplained ecological relationships, or encounter environmental data they cannot interpret. When appropriately calibrated and supported, this anxiety functions as an adaptive response that motivates information-seeking behavior [19]. The distinction between productive and debilitating anxiety is crucial. Productive learning anxiety occurs within an optimal range sufficient to create awareness of knowledge gaps without overwhelming cognitive resources or triggering avoidance behaviors [18]. When learning environments provide adequate support structures, this moderate anxiety can be channeled into constructive inquiry rather than escalating into disengagement.

2.5. The anxiety-curiosity nexus: theoretical foundations

Information gap theory provides the foundational framework for understanding anxiety's role in curiosity activation [20]. The theory posits that curiosity arises from recognition of a gap between what one knows and wants to know, with this gap creating an aversive state that motivates inquiry. Learning anxiety represents the affective dimension of this gap awareness, the emotional discomfort accompanying cognitive recognition of insufficient knowledge. Critically, moderate information gaps produce optimal curiosity: gaps too small fail to generate sufficient motivation, while gaps too large may induce inadequacy that suppresses inquiry. Empirical research supports this relationship. Studies demonstrate that inducing awareness of knowledge gaps, thereby creating moderate learning anxiety significantly enhances

epistemic curiosity and subsequent learning performance [21]. Deliberately designed information gaps improve students' state curiosity, knowledge retention, and achievement, with effects persisting across grade levels [6].

The relationship between learning anxiety and curiosity differs systematically by curiosity type. Deprivation-type (D-type) curiosity, characterized by tension and the compulsive drive to reduce cognitive dissonance, aligns closely with the discomfort inherent in learning anxiety [3]. When students experience anxiety about knowledge gaps encountering an unidentifiable organism or unexplained ecological pattern, this tension directly fuels D-type curiosity, driving focused information acquisition. However, the relationship with interest-type (I-type) curiosity is more complex and conditional. I-type curiosity, characterized by pleasurable anticipation of discovery, typically requires psychological safety and autonomy [2]. Excessive anxiety may inhibit I-type curiosity by consuming cognitive resources and reducing perceived competence [19]. Conversely, when learning anxiety is moderate and accompanied by supportive scaffolding such as well-designed AR exploration tools, it may facilitate I-type curiosity through a two-stage process. First, manageable anxiety triggers D-type curiosity, prompting information-seeking. Second, as learners successfully fill knowledge gaps through discovery, anxiety reduction creates psychological space for I-type curiosity to emerge. Competence gained through anxiety resolution can transform obligatory information-seeking into intrinsically motivated exploration, particularly when the learning environment provides autonomy-supportive features.

Digital technologies, particularly augmented reality with IoT sensors, offer unique affordances for strategically inducing productive learning anxiety while minimizing excessive stress. AR-IoT systems can create controlled information gaps by presenting intriguing but incomplete ecological data, highlighting discrepancies between predictions and observations, or revealing hidden relationships. The immediacy and contextual relevance of AR-delivered prompts amplify their productive potential: because AR overlays information onto students' physical surroundings, knowledge gaps revealed are directly tied to their lived environment rather than abstract phenomena, increasing personal relevance [14]. Critically, these technologies also provide scaffolding to channel anxiety into productive curiosity. Progressive hints, visualization tools, and immediate sensor feedback validate discoveries, reinforce competence, and facilitate the anxiety-to-interest transition. This combination creates conditions conducive to both curiosity types.

Empirical evidence for the pedagogically productive role of anxiety-induced knowledge gaps in technology-enhanced science education is provided by Lammina and Chase [22], who conducted a randomized experiment comparing two versions of an interactive biology simulation: one presenting complete taxonomic information upfront, and one introducing organisms through progressively revealed classification puzzles designed to induce moderate uncertainty. Students in the uncertainty condition reported significantly higher learning anxiety at mid-activity, yet simultaneously demonstrated greater epistemic curiosity, deeper information-seeking behaviors, and superior post-test performance on transfer tasks. Critically, the anxiety-to-curiosity transition was fully mediated by students' perception that the environment was controllable and that errors were informative rather than evaluative. This finding converges with the present study's rationale for embedding IoT-driven real-time feedback into the AR exploration environment, ensuring that students' initial mosquito-related anxiety encounters a scaffolded, competence-affirming response rather than unstructured uncertainty.

Based on this foundation, we propose:

H1. Learning anxiety will positively influence deprivation-type curiosity (D-type) in AR-enhanced environmental education.

H2. Learning anxiety will positively influence interest-type curiosity (I-type) in AR-enhanced environmental education.

2.6. Integrating anxiety, curiosity, and flow: toward a unified framework

Having established the theoretical foundations of learning anxiety, curiosity, and flow independently, this section synthesizes these mechanisms into an integrated framework explaining the complete motivational trajectory in AR-enhanced environmental education. This integration addresses a critical gap: while studies have examined curiosity-flow relationships [23] and anxiety-curiosity connections [2] separately, few have investigated how these processes interact sequentially to produce sustained engagement in technology-enhanced learning.

The proposed framework posits a three-stage sequence that transforms initial discomfort into lasting commitment: (1) learning anxiety triggers curiosity-driven information seeking, (2) curiosity facilitates flow through focused exploration, and (3) flow enhances continuance intention through intrinsically rewarding experiences. This addresses the dual challenges in elementary environmental education: overcoming local biodiversity's "invisibility problem" while sustaining engagement beyond initial encounters.

The temporal dynamics reflect learning motivation's progressive nature. Initial anxiety serves as the catalyst when students might otherwise remain passive observers. As exploration resolves information gaps, conditions emerge for flow: clear goals, immediate feedback from discoveries, and appropriate challenge-skill balance. The positive effect generated during flow then motivates continued platform use, completing the cycle from tension to sustained engagement.

The synergy between curiosity and flow suggests multiplicative rather than additive effects. Both mechanisms share emphasis on intrinsic motivation and focused attention: curiosity directs cognitive resources toward specific targets creating the concentrated attention central to flow states [9]. When digital environments provide immediate feedback on discovery progress, they satisfy both curiosity's need for information closure and flow's requirement for performance monitoring. A key insight concerns challenge-skill balance management. While flow requires perceived challenges to match perceived skills [9], curiosity theory provides stabilizing mechanisms: D-type curiosity reframes difficulty as resolvable information gaps rather than competence threats, while I-type curiosity maintains engagement through intrinsically rewarding exploration even when progress is incremental. This creates tolerance for challenge that prevents anxiety (distinct from productive learning anxiety) and boredom.

AR-IoT systems operationalize this framework through affordances simultaneously addressing multiple motivational mechanisms. Location-based augmentation creates personally relevant information gaps in students' immediate environments. Real-time sensor feedback satisfies information closure needs while providing performance monitoring. Adaptive difficulty calibration, adjusting ecological complexity based on demonstrated competence, maintains challenge-skill equilibrium essential for flow. These affordances enable personalization difficult to achieve traditionally. By tracking individual progress, systems can calibrate anxiety induction to ensure optimally-sized information gaps, adjust challenges to maintain flow-conducive balance, and provide precisely-timed scaffolding. This addresses a fundamental limitation: optimal anxiety levels, curiosity triggers, and flow conditions vary across learners based on prior knowledge and motivational profiles.

Despite demonstrated advantages of curiosity-based instruction [6] and flow-enhancing design [5], most studies examine these phenomena separately. Few interventions have deliberately designed environments operationalizing the complete anxiety-curiosity-flow sequence, and fewer still have tested whether hypothesized sequential relationships hold in authentic educational contexts. Moreover, research rarely examines how technological affordances support psychological transitions, particularly the D-to-I curiosity shift and subsequent flow emergence.

2.7. Research hypotheses

Within this integrated framework, our hypotheses test the complete motivational sequence:

H1–H2 (Initiation). Learning anxiety triggers both deprivation-type curiosity (D-type) and interest-type curiosity (I-type) in AR-enhanced contexts.

H3–H4 (Sustain). Both D-type and I-type curiosity facilitate flow through focused exploration.

H5 (Outcome). Flow experience enhances continuance intention for sustained engagement.

By empirically testing this sequence with elementary students in AR-enhanced dengue vector education, the study addresses the identified gap and provides evidence for the theoretical integration of anxiety, curiosity, and flow in technology-mediated environmental learning. Findings will inform both theoretical understanding of motivational dynamics and practical design principles for creating environmental learning experiences that achieve immediate engagement and long-term persistence.

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3. Methods

3.1. Research design

This study employed a theory-driven design approach to develop and validate an AR-enhanced environmental education curriculum grounded in integrated curiosity-flow theory. The research followed a sequential logic: theoretical framework construction → curriculum design → implementation → psychological response measurement → theoretical validation.

We adopted a single-group design rather than a controlled experimental design for three methodological reasons. First, as an initial validation of the integrated curiosity-flow framework in AR-enhanced environmental education, this study prioritizes establishing whether the theoretically-predicted psychological mechanisms can be triggered through digital technology design. Second, the research originated from a funded project, which required developing and evaluating a complete instructional system rather than isolating individual variables through experimental manipulation. Third, structural equation modeling allows examination of complex relationships among latent psychological constructs, providing insights into mediating processes that simple group comparisons cannot reveal. While future research should employ controlled designs to establish causal effects relative to traditional instruction, this study's contribution lies in demonstrating the feasibility and psychological validity of the integrated theoretical framework.

3.2. AR application design and implementation

The AR application was developed to directly operationalize the integrated curiosity-flow framework, with each design feature mapped to a specific motivational stage. Built on Unity with the Vuforia AR SDK and deployed on iOS and Android, the App streamed real-time IoT sensor data, including standing water volume and humidity readings, as AR overlays when students scanned pre-deployed trigger markers across the campus.

The theory-design correspondence was implemented across three stages. To induce D-type curiosity, the App presented deliberately incomplete environmental cues at the task interface, withholding

complete prevention guidance to generate cognitive knowledge gaps consistent with Loewenstein's information gap theory [20]. To facilitate transformation toward I-type curiosity, students autonomously selected exploration paths, with animated reward effects and cumulative scores upon task completion rendering discovery intrinsically pleasurable [2]. To support flow, a difficulty-tiered task sequence ensured continuous challenge–skill recalibration, while clear objectives and immediate AR visual feedback satisfied the three preconditions identified by Csikszentmihalyi: clear goals, immediate feedback, and challenge–skill equilibrium [9].

At the implementation level, the experience comprised three phases: task assignment, campus exploration, and outcome reporting. Students received randomly assigned lists of seven breeding-site targets with minimal cues and no precise location information. Working in small groups, they scanned AR markers to reveal real-time ecological overlays, with each discovery automatically logged. Students then completed reflective prompts within the App and submitted findings for teacher-facilitated discussion. This three-phase structure ensured all participants experienced the complete motivational trajectory within an ecologically valid context.

Table 1 presents the complete technical specifications of the AR–IoT system deployed in this study. The application was built using Unity 2020 LTS integrated with Vuforia Engine SDK 9.4.6, selected for its robust image-target recognition performance in outdoor campus environments with variable lighting conditions. The AR interface was deployed on school-issued iOS and Android tablets, ensuring consistency of display parameters across participating classrooms. IoT environmental monitoring was implemented through DHT22 temperature–humidity modules and ultrasonic water-level sensors installed at seven designated campus hotspot locations identified through prior field surveys as high-risk mosquito breeding areas. Sensor data were transmitted in real time via an MQTT protocol over the school Wi-Fi network and logged to a Firebase Realtime Database, enabling the AR application to retrieve and overlay live environmental readings onto the physical campus scene. All AR trigger markers consisted of Vuforia Image Targets printed at a standardized 10 cm × 10 cm format and laminated for outdoor durability. To support replication, the complete trigger marker image set, sensor configuration files, and Unity project scene files are archived and available upon reasonable request to the corresponding author.

The three-layer architecture of the AR–IoT learning system illustrated in Fig. 1. The physical layer comprises IoT environmental sensors, specifically DHT22 temperature–humidity modules and ultrasonic water-level detectors, deployed at eight campus hotspot locations and transmitting real-time data via MQTT protocol to a Firebase cloud backend. The digital layer consists of the Unity/Vuforia-based mobile application, which retrieves sensor readings from the cloud and renders them as AR

information overlays upon recognition of printed trigger markers distributed across the campus. The pedagogical layer maps each technological component to a specific stage in the motivational sequence: sensor-generated data gaps are deliberately withheld to sustain D-type curiosity; the free-navigation task structure and autonomous marker-scanning mechanism support the emergence of I-type curiosity; and real-time animated achievement feedback, combined with progressive difficulty calibration, creates the challenge–skill balance conditions necessary for flow. This architecture ensures that the theoretical constructs operationalized in the measurement model correspond directly to identifiable features of the learning system, thereby supporting the construct validity of the study's structural equation modeling approach (Fig. 2).

Panel (a) shows the gamified mission interface, which presents a series of levels that each pose a different mosquito-related challenge; the running score and the upload function for submitting fieldwork provide immediate, tangible feedback, and each new challenge deliberately surfaces gaps in students' prior knowledge, generating the knowledge-gap tension characteristic of D-type curiosity. Panel (b) shows an augmented reality field task in which students search the physical campus and inspect candidate breeding sites to identify disease-carrying larvae smaller than 8 mm; this self-directed, question-driven exploration, where learners must reason about habitat clues before confirming a target, characterizes the I-type curiosity facilitation stage, shifting engagement from anxiety-driven information seeking toward intrinsically motivated discovery. Panel (c) shows a second augmented reality task in which students capture dengue-transmitting mosquitoes; the animated reinforcement and cumulative progress feedback accompanying a successful capture draw on principles of competence validation and progress visualization to sustain engagement and support the transition into flow experience.

Fig. 3 presents the campus deployment configuration together with the system in authentic operation, illustrated through a representative school. Panel (a) shows the spatial deployment configuration. Each school's outdoor spaces were divided into five designated exploration zones, namely the main courtyard, the building-perimeter walkways, the planted greenery areas, and the drainage corridor, chosen on the basis of prior field surveys confirming their status as authentic high-risk mosquito-breeding environments under southern Taiwan's subtropical conditions. Within each zone, AR trigger markers were placed at positions presenting ecologically meaningful visual cues (e.g., near standing-water receptacles, beneath dense foliage, or adjacent to drainage infrastructure), while IoT sensor modules were co-located at the same points to provide the real-time environmental data that the AR overlay retrieves and displays. The spatial distribution was designed so that students navigating the campus freely would encounter approximately equal numbers of target sites per zone, maintaining consistent challenge levels across exploration paths. Standardized across all participating schools through pre-session site visits by the research team, this configuration constitutes a replicable environmental scaffold for the pedagogical intervention.

Panels (b) to (d) show how the AR–IoT application mediates students' interaction with the physical campus across the two-session curriculum. The interaction follows a scan-recognize-overlay sequence: a student scans a printed trigger marker (b), the Vuforia engine recognizes the image target, and the application superimposes ecological content and live sensor data onto the physical scene in real time. Panel (c) shows this overlay as seen by the user, including the live IoT readings (ambient temperature and standing-water depth) that make otherwise invisible environmental conditions perceptible at each breeding-site location, and panel (d) shows students in collaborative exploration during class. This design operationalizes the study's motivational logic: the deliberately incomplete information presented at the start of each mission creates the knowledge-gap tension characteristic of D-type curiosity; the question-driven, self-directed search for markers and the progressive revelation of real-time ecological data sustain intrinsically motivated I-type

Table 1
AR–IoT system technical specifications.

Component	Specification
Development Engine	Unity 2020 LTS
AR Framework	Vuforia Engine SDK 9.4.6
Target Platforms	iOS 14+ / Android 10+
IoT Protocol	MQTT over Wi-Fi
Sensor Types	DHT22 temperature–humidity + ultrasonic water-level detectors
Cloud Backend	Firebase Realtime Database
AR Trigger Markers	Vuforia Image Targets, 10 cm × 10 cm, laminated for outdoor use
Microcontroller	Arduino Uno R3
Accessibility	Font size ≥ 14 pt; high-contrast AR overlays; Mandarin Chinese interface

Note. The complete trigger marker image set, sensor configuration files, and Unity project scene files are archived and available upon reasonable request to the corresponding author.

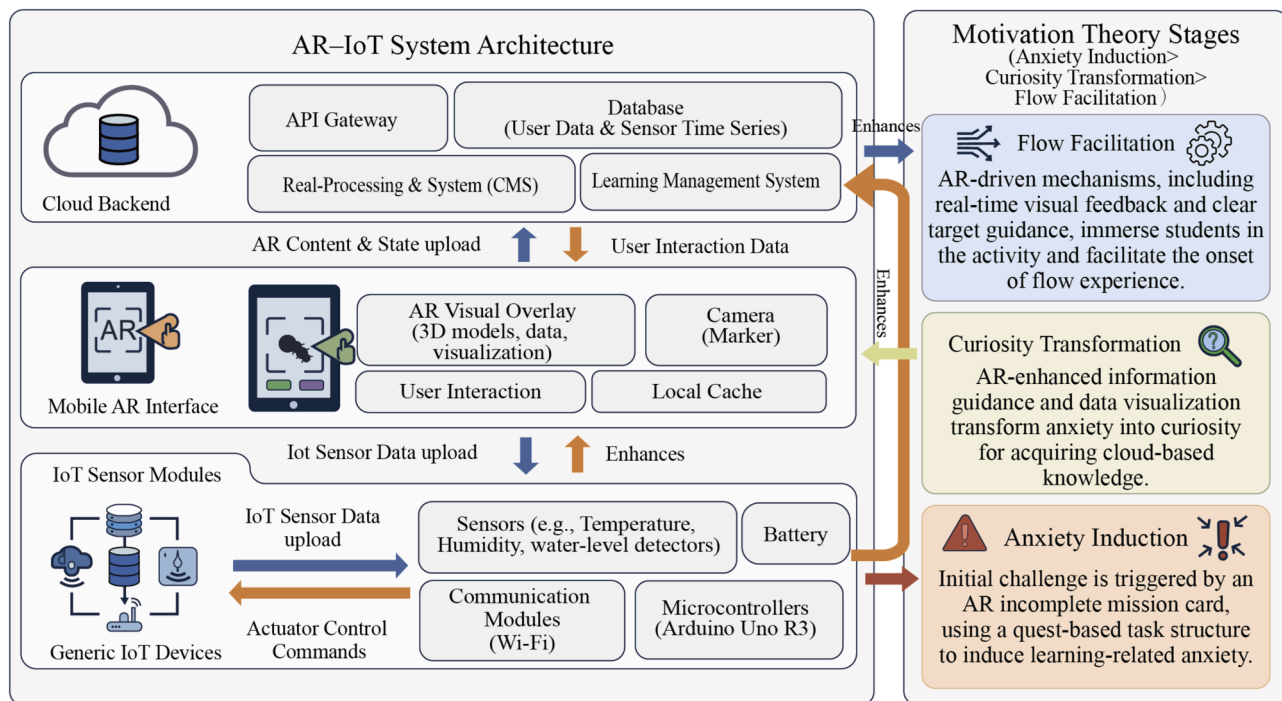


Fig. 1. AR-IoT system architecture. Note. AR-IoT system architecture illustrating the relationships among the mobile AR interface, IoT sensor modules, cloud backend, and the three motivational stages (anxiety induction → curiosity transformation → flow facilitation) each component is designed to support.

exploration; and the achievement feedback following successful site identification supports the transition into flow experience.

3.3. Participants and procedure

Participants were recruited from elementary schools in Taiwan through partnerships with local health departments conducting dengue fever prevention programs. A purposeful sampling technique was used to ensure participants met specific developmental and curricular criteria. Fifth-grade students were selected based on two criteria: (1) According to Taiwan's elementary science curriculum guidelines, biological content is formally introduced in third grade, ensuring students possessed sufficient foundational knowledge to understand dengue fever vector control principles; (2) This age group demonstrates adequate cognitive abilities to operate digital learning tools while being developmentally appropriate for curiosity-driven instruction [24]. Participation was voluntary, and parental consent was obtained for all students prior to the study. The initial sample included 520 fifth-grade students. After accounting for absences during the intervention period and incomplete questionnaire responses, 423 students (205 male, 218 female) provided usable data, yielding an effective response rate of 81.3%. The mean age of participants was 11.2 years ($SD = 0.45$). All participants were enrolled in public elementary schools in southern Taiwan, regions where dengue fever prevention is an authentic public health concern.

Participant recruitment was coordinated by the co-principal investigator of the research project through formal partnerships with local health departments in southern Taiwan. Health department personnel issued invitations to eligible public elementary schools, with science teachers at each school serving as on-site liaisons responsible for scheduling and student communication. Schools confirmed participation only after verifying that the two-session intervention could be accommodated within their regular science curriculum periods, ensuring that the study context aligned with routine instructional practices while addressing local dengue fever prevention priorities.

The AR-based learning activities were conducted entirely within the students' own school campuses, requiring no displacement to external

venues. Each participating school designated specific campus zones for the exploration tasks, encompassing areas such as building courtyards, corridor-adjacent drainage channels, planted greenery zones, and building perimeters where locations represent common *Aedes* mosquito breeding habitats in southern Taiwan's subtropical school environments. Prior to each session, members of the research team visited each school to collaborate with teachers on deploying AR trigger markers and conducting system functionality checks, ensuring operational consistency across all participating sites. Conducting the activities within students' familiar campus environments reinforced the contextual relevance of the AR-overlaid ecological information and reduced extraneous cognitive load, thereby creating conditions conducive to the authentic emergence of the proposed motivational mechanisms.

Data collection occurred during the 2021 academic year. The intervention was administered during regular science class across two 40-minute periods.

Session 1: Teachers presented dengue fever epidemiological data, mosquito habitat ecology, and breeding prerequisites, while deliberately leaving prevention strategies incomplete. The session concluded by introducing the AR app interface and challenge rules.

Session 2: Students used the AR app to search for mosquito breeding sites on campus and completed gamified environmental cleanup missions in small groups.

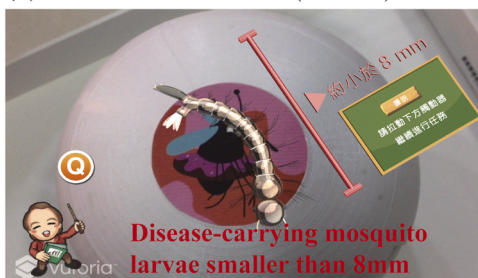
Immediately following Session 2, students completed a structured questionnaire assessing learning anxiety, curiosity dimensions, flow experience, and continuance intention. Questionnaire administration was supervised by classroom teachers and research assistants to ensure comprehension and minimize missing data. Students were assured that responses would not affect academic grades and that participation was voluntary. Questionnaire completion took approximately 20–25 min.

All procedures were approved by the institutional review board and adhered to ethical guidelines for research involving minors. Informed consent was obtained from parents/guardians, and student assent was obtained prior to participation.

(a) Game mission interface



(b) AR larva identification (< 8 mm)



(c) AR mosquito capture



Fig. 2. Annotated app interface screenshots. Panel (a) the gamified mission interface, which presents a series of levels that each pose a different challenge, shown with a score display and an upload function for submitting fieldwork; (b) an augmented reality field task for identifying disease-carrying larvae smaller than 8 mm; and (c) an augmented reality field task for capturing dengue-transmitting mosquitoes.

3.4. Measurement instruments

Entirely measurement instruments employed validated scales that were linguistically adapted to ensure age-appropriate comprehension for elementary students while preserving the underlying theoretical constructs. All instruments utilized five-point Likert scales ranging from 1 (strongly disagree) to 5 (strongly agree).

The study assessed epistemic curiosity using Litman and Jimerson's [25] two-dimensional scale, which distinguishes between D-type curiosity (deprivation sensitivity) and I-type curiosity (interest). The five D-type items captured knowledge-gap driven exploration (e.g., "I felt I absolutely needed to learn about mosquito breeding sites"), while the five I-type items measured intrinsic enjoyment of discovery (e.g., "I enjoyed discovering where mosquitoes hide on campus"). Both dimensions demonstrated excellent internal consistency (D-type $\alpha = .899$; I-type $\alpha = .908$). Flow experience was measured using an 11-item scale adapted from Csikszentmihalyi [9], which assessed core flow dimensions including challenge-skill balance, clear goals, immediate feedback, and concentration (e.g., "I was so focused on the activity that I forgot about time"; $\alpha = .931$). Students' continuance intention was evaluated through a six-item scale adapted from [26], measuring their willingness to engage in future environmental education activities using similar technological approaches (e.g., "I want to learn more about environmental protection using similar tools"; $\alpha = .884$). Finally,

mosquito-related learning anxiety capturing cognitive, affective, and behavioral anxiety dimensions related to dengue fever knowledge and prevention (e.g., "I worried about not knowing enough about dengue fever prevention"; $\alpha = .872$).

Data analysis employed structural equation modeling (SEM) using AMOS software (Version 29.0), selected for its capacity to simultaneously examine measurement quality and structural relationships among latent constructs. The confirmatory factor analysis model is presented in Fig. 4. The analysis followed a two-stage approach, beginning with confirmatory factor analysis (CFA) to validate the measurement model and ensure that observed indicators adequately represented their respective latent constructs. Reliability was assessed through Cronbach's alpha coefficients and composite reliability (CR), with values exceeding .70 considered acceptable. Convergent validity was evaluated through standardized factor loadings (threshold: $> .50$) and average variance extracted (AVE $> .50$), while discriminant validity was examined by comparing the square root of AVE with inter-construct correlations. The structural model was then tested to examine hypothesized relationships among constructs, with model fit evaluated using multiple indices: chi-square to degrees of freedom ratio (χ^2/df), root mean square error of approximation (RMSEA), comparative fit index (CFI), Tucker-Lewis's index (TLI), and parsimony goodness-of-fit index (PGFI). Following established guidelines [27,28], acceptable fit was indicated by $\chi^2/df < 3.0$, RMSEA $< .08$, and CFI and TLI values $> .90$.

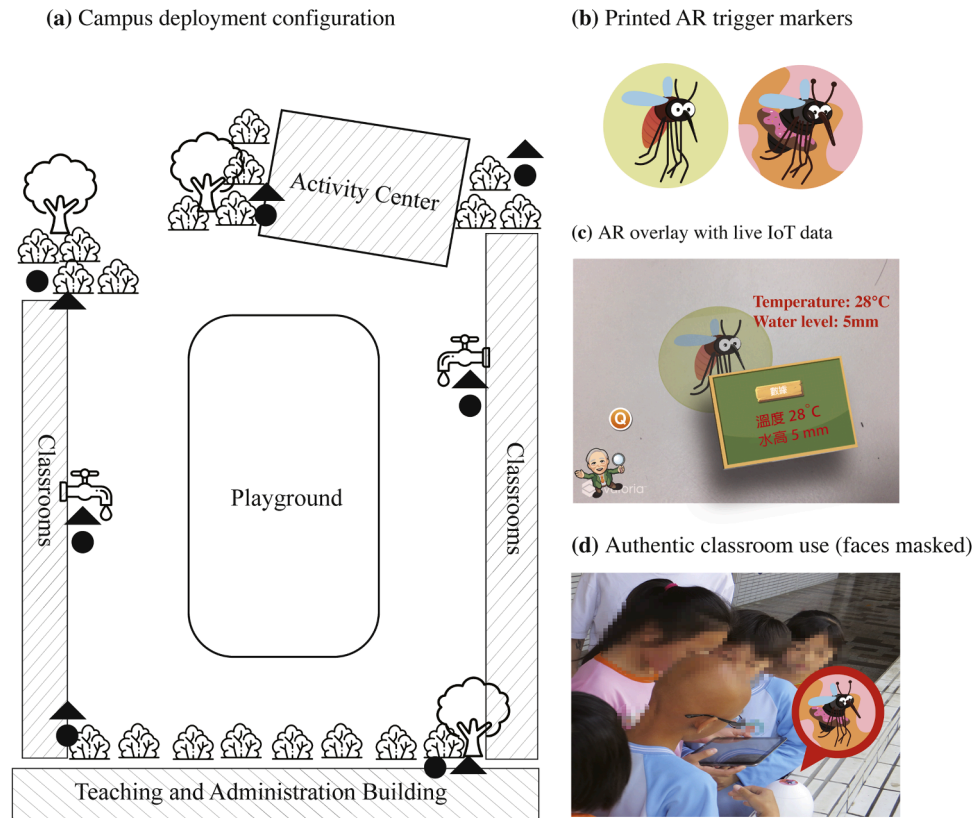


Fig. 3. Campus deployment and authentic operation of the AR-IoT system. Note. (a) Schematic of the campus deployment configuration, showing the spatial distribution of AR trigger markers (▲) and IoT sensor modules (●) across the designated exploration zones of a representative participating school. (b) Examples of the printed AR trigger markers (laminated Vuforia image targets), whose high-feature-density, child-friendly designs support robust recognition under variable outdoor lighting. (c) The AR overlay as seen by the user: upon recognition, the application superimposes ecological content together with live IoT sensor readings (ambient temperature 28 °C and standing-water depth 5 mm) onto the physical marker; the original Chinese interface text is translated in red for readers. (d) Students engaged in collaborative campus exploration during Session 2; participant faces have been masked to protect privacy.

Mediating effects were examined using bootstrapping procedures with 5000 resamples and 95% bias-corrected confidence intervals, with mediation considered significant when the indirect effect's confidence interval did not include zero. All significance tests employed a two-tailed alpha level of .05.

4. Results

4.1. Measurement model assessment

The measurement model demonstrated robust psychometric properties across all constructs. Internal consistency reliability was excellent for all measures, with Cronbach's alpha coefficients ranging from .872 (mosquito-related anxiety) to .931 (flow experience), all exceeding the recommended .70 threshold. Composite reliability values paralleled these findings, ranging from .872 to .930, further confirming measurement stability.

Convergent validity was established through two complementary approaches. First, all standardized factor loadings exceeded the .50 threshold, ranging from .665 to .851, indicating that observed items adequately represented their underlying constructs. Second, average variance extracted (AVE) values demonstrated that all five constructs surpassed or approached the .50 criterion: I-type curiosity (AVE = .666), D-type curiosity (AVE = .641), continuance intention (AVE = .562), flow experience (AVE = .550), and mosquito-related anxiety (AVE = .533). Complete measurement model indices are presented in Table 2.

Examination of construct means revealed theoretically meaningful patterns. Students reported highest levels of continuance intention ($M = 3.97$, $SD = .98$) and I-type curiosity ($M = 3.93$, $SD = .89$), suggesting

strong engagement and intrinsic motivation following the curriculum. Flow experience exhibited the lowest mean ($M = 2.83$, $SD = .83$), indicating moderate immersion levels that may benefit from further instructional refinement. D-type curiosity ($M = 3.40$, $SD = .86$) fell between the two curiosity types, consistent with theoretical expectations of its role as a transitional motivational state bridging knowledge gaps and intrinsic interest.

4.2. Correlational analysis

Bivariate correlations among the five latent constructs revealed systematic relationships consistent with theoretical expectations (see Table 3). All correlations were statistically significant at the $p < .001$ level, providing initial support for the hypothesized structural model.

The strongest association emerged between I-type curiosity and continuance intention ($r = .656$), supporting the theoretical proposition that interest-driven exploration serves as a primary driver of sustained engagement in environmental education. This finding aligns with curiosity-flow theory's emphasis on intrinsic motivation as a prerequisite for long-term behavioral commitment. The two curiosity dimensions exhibited substantial intercorrelation ($r = .552$), confirming Litman's [2] conceptualization of D-type and I-type curiosity as related yet distinct motivational states rather than opposing constructs.

Flow experience demonstrated robust associations with all variables, particularly with mosquito-related anxiety ($r = .587$) and both curiosity types (I-type: $r = .554$; D-type: $r = .534$). The strong anxiety-flow relationship suggests that moderate anxiety induction through personally relevant health threats may enhance immersive learning by heightening attentional focus and perceived task significance. Notably, anxiety

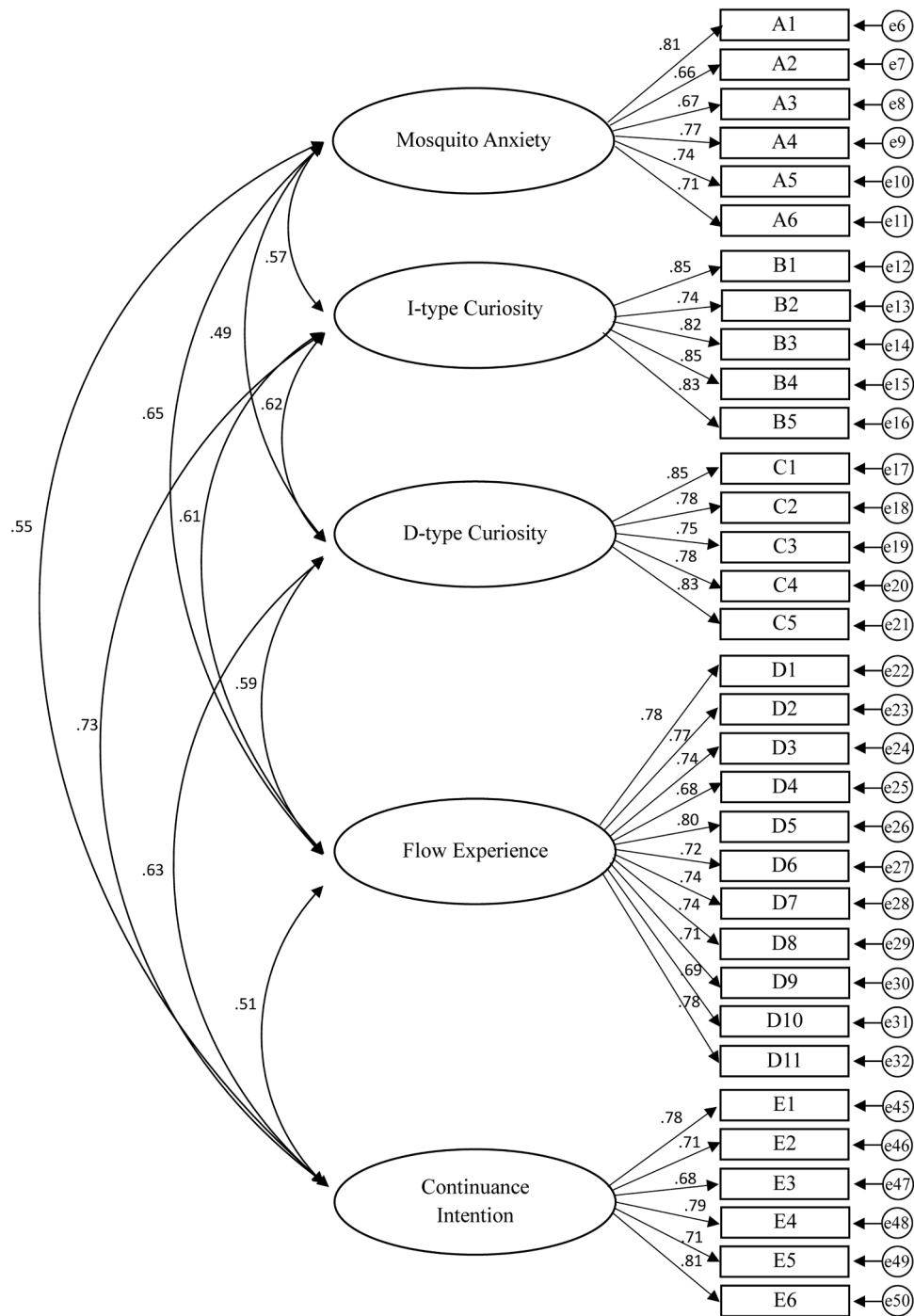


Fig. 4. Confirmatory factor analysis model. Note. All factor loadings significant at $p < .001$. Curved arrows represent correlations between latent constructs. Model fit: $\chi^2/df = 2.122$, $RMSEA = .052$, $GFI = .878$, $CFI = .938$, $TLI = .933$.

correlated more strongly with I-type curiosity ($r = .509$) than with D-type curiosity ($r = .430$), implying that anxiety alone may be insufficient to trigger knowledge-gap motivation without supportive instructional scaffolding that explicitly frames health threats as remediable through learning.

4.3. Structural model evaluation

The hypothesized structural model demonstrated acceptable to good fit across multiple evaluation criteria. (see Fig. 4) The chi-square to degrees of freedom ratio ($\chi^2/df = 2.640$) fell below the recommended 3.0 threshold, while $RMSEA$ (.062) indicated good fit, below the .08

criterion [27]. Incremental fit indices exceeded or approached recommended thresholds ($TLI = .901$, $CFI = .909$), demonstrating strong model performance. GFI (.852) and $AGFI$ (.830) remained within acceptable ranges for complex models ($> .80$). The parsimonious fit index ($PGFI = .741$) suggested good model parsimony. Collectively, these indices supported the theoretical framework's structural validity.

All hypothesized paths achieved statistical significance at $p < .001$ (see Table 4 & Fig. 5). Mosquito-related anxiety significantly predicted both I-type curiosity ($\beta = .62$, $t = 10.72$) and D-type curiosity ($\beta = .54$, $t = 9.52$), accounting for 38% and 29% of variance respectively. The stronger effect on I-type curiosity suggests that anxiety-inducing contexts more effectively trigger interest-based exploration than

Table 2
Construct-level reliability and validity indices.

Construct	M	SD	α	CR	AVE	Items	Factor Loading Range
Mosquito Anxiety	3.66	.84	.872	.873	.533	6	.665-.810
I-type Curiosity	3.93	.89	.908	.909	.666	5	.736-.849
D-type Curiosity	3.40	.86	.899	.899	.641	5	.753-.851
Flow Experience	2.83	.83	.931	.930	.550	11	.677-.803
Continuance Intention	3.97	.98	.884	.884	.562	6	.683-.809

Note. $N = 423$. M = Mean; SD = Standard Deviation; α = Cronbach's Alpha; CR = Composite Reliability; AVE = Average Variance Extracted.

knowledge-gap tension when appropriate technological affordances support active discovery.

Both curiosity dimensions significantly predicted flow experience, with I-type curiosity exerting stronger effects ($\beta = .44$, $t = 8.95$) compared to D-type curiosity ($\beta = .38$, $t = 7.93$). Together, the two curiosity types explained 52% of variance in flow experience, indicating that epistemic motivation particularly interest-driven exploration substantially contributes to optimal learning states. Flow experience, in turn, significantly predicted continuance intention ($\beta = .53$, $t = 9.67$), accounting for 28% of variance and confirming that immersive learning experiences enhance sustained engagement with environmental education.

Bootstrap analysis with 5000 resamples confirmed significant indirect effects through the integrated curiosity-flow pathway. The indirect pathway from mosquito-related anxiety to continuance intention through I-type curiosity and flow experience yielded a significant effect of .143 (95% CI [.085, .220], $p = .001$). The parallel pathway through D-type curiosity produced an indirect effect of .109 (95% CI [.065, .170], $p = .001$). These findings demonstrate that both curiosity dimensions mediate the relationship between initial anxiety induction and sustained behavioral engagement, with the I-type pathway exhibiting slightly stronger mediation.

Collectively, the structural model successfully validated the integrated curiosity-flow theoretical framework, with all hypothesized paths achieving significance and the model explaining substantial variance in flow experience (52%) and moderate variance in continuance intention (28%). The stronger effects of I-type curiosity on both flow and continuance intention underscore the importance of transforming initial knowledge-gap tension into pleasurable exploration through appropriate technological scaffolding, supporting the proposition that AR-enhanced environmental education can systematically trigger psychological processes from anxiety induction through curiosity transformation to sustained engagement.

5. Discussion

This study demonstrates that AR-enhanced environmental education grounded in integrated curiosity-flow theory can systematically trigger psychological processes that promote sustained engagement. All hypothesized paths achieved significance, with the model explaining 52% of variance in flow experience and 28% in continuance intention. These

Table 4
Standardized path coefficients and model estimates.

Hypothesis	Path	β	SE	t	R^2
H1	Mosquito Anxiety $\rightarrow$ I-type Curiosity	.615***	.077	10.72	.38
H2	Mosquito Anxiety $\rightarrow$ D-type Curiosity	.540***	.059	9.52	.29
H3	D-type Curiosity $\rightarrow$ Flow Experience	.380***	.050	7.93	—
H4	I-type Curiosity $\rightarrow$ Flow Experience	.439***	.040	8.95	.52†
H5	Flow Experience $\rightarrow$ Continuance Intention	.529***	.060	9.67	.28

*Note. $N = 423$. β = standardized path coefficient; SE = standard error; R^2 = variance explained in endogenous variable. †Combined effect of H3 and H4. *** $p < .001$.

findings illuminate how digital technologies provide unique affordances for operationalizing motivational theories in educational practice.

5.1. Anxiety as a trigger for both curiosity types

Mosquito-related anxiety significantly predicted both D-type ($\beta = .54$) and I-type curiosity ($\beta = .62$), demonstrating successful implementation of staged curiosity induction. Critically, anxiety exhibited stronger effects on I-type curiosity, suggesting that when anxiety-inducing contexts are coupled with appropriate exploration tools, learners more readily engage in interest-driven exploration than knowledge-gap closure alone. This pattern contrasts with pure information-gap approaches that may sustain tension without facilitating pleasurable discovery [6].

The AR exploration system provided technological affordances that supported this transformation through interconnected mechanisms: real-time visual feedback through colored trigger tags enabled immediate connection between physical discovery and digital information [29], autonomous navigation through randomized task assignment gave students control over exploration paths, and achievement visualization through animated feedback reinforced transformation by providing tangible evidence of learning progress [14]. These affordances are difficult to replicate in non-digital environments, traditional hands-on exploration lacks the precision timing and adaptive scaffolding that digital systems provide, while purely digital environments miss the embodied engagement that physical manipulation offers.

5.2. Curiosity types differentially predict flow experience

Both curiosity types significantly predicted flow experience, with I-type showing stronger effects ($\beta = .44$) than D-type ($\beta = .38$). This finding supports the theoretical proposition that curiosity and flow represent complementary motivational mechanisms [2]: curiosity drives information seeking, while flow maintains prolonged engagement through optimal challenge-skill balance.

The moderate flow levels observed ($M = 2.83$) suggest room for improvement. This likely reflects inherent challenges in creating flow during initial exposure to new technologies, our participants were novice AR users navigating both technological interfaces and scientific

Table 3
Inter-construct correlations.

	Mosquito Anxiety	I-type Curiosity	D-type Curiosity	Flow Experience	Continuance Intention
Mosquito Anxiety	—				
I-type Curiosity	.509***	—			
D-type Curiosity	.430***	.552***	—		
Flow Experience	.587***	.554***	.534***	—	
Continuance Intention	.482***	.656***	.568***	.472***	—

Note. *** $p < .001$.

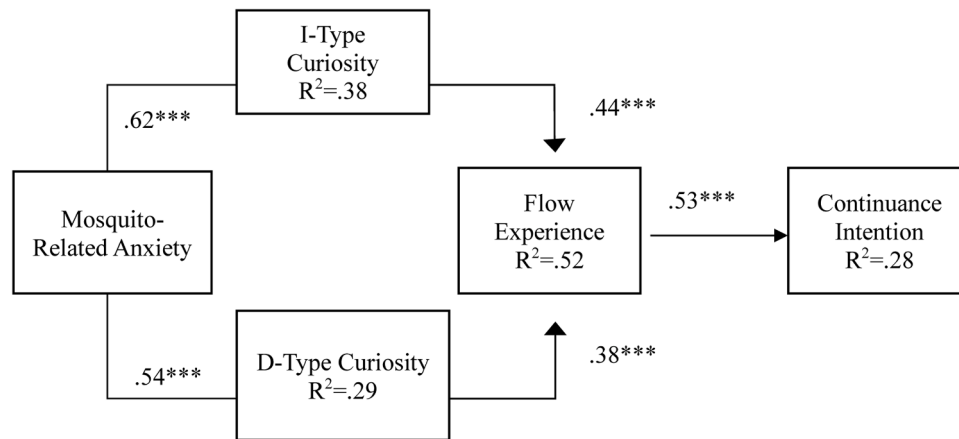


Fig. 5. Structural model with standardized path coefficients. Note. All paths significant at $***p < .001$. Model fit: $\chi^2/df = 2.640$, $RMSEA = .062$, $TLI = .901$, $CFI = .909$.

content simultaneously. The significant flow-continuance relationship ($\beta = .53$) indicates that even moderate flow experiences substantially influenced sustained engagement, but future iterations should incorporate extended practice opportunities to allow skill development and deeper immersion.

5.3. Flow experience promotes continuance intention

Flow experience significantly predicted continuance intention ($\beta = .53$, $R^2 = .28$), confirming that optimal learning states promote long-term engagement with educational technologies. This relationship has important practical implications, as environmental education programs often struggle with maintaining student participation beyond initial exposure [29,30]. By designing for flow through curiosity-driven exploration, educators can create experiences that learners intrinsically want to continue.

The mediation analysis revealed that both curiosity pathways contributed to sustained engagement through flow experience, with the I-type pathway showing slightly stronger indirect effects (.143 vs. .109). This underscores the importance of transforming initial anxiety into pleasurable exploration rather than merely resolving knowledge gaps. However, the modest variance explained in continuance intention (28%) indicates that factors beyond immediate flow experience such as perceived relevance, social norms, and institutional support influence sustained engagement.

5.4. Implications

This study offers both theoretical and practical implications for educational technology. Theoretically, it demonstrates the value of integrating curiosity and flow into a unified motivational framework, providing empirical evidence for their sequential interaction in AR-enhanced learning. It also clarifies how specific affordances such as real-time feedback, learner autonomy, and achievement visualization map onto psychological mechanisms, and extends curiosity theory into environmental education by showing the robustness of the anxiety–curiosity–flow pathway in a health-related ecological context.

Practically, the findings suggest design principles for sustaining motivation in AR-enhanced curricula. Designers should strategically introduce information gaps to trigger curiosity, while pairing them with supportive autonomy and clear goals to prevent confusion. Real-time, tangible feedback is critical for validating learner discoveries and maintaining engagement. Finally, adaptive progression of challenge levels ensures that the pathway from initial anxiety through curiosity to flow can be sustained over extended experiences.

5.5. Design principles grounded in empirical finding

Beyond these general principles, the empirical path coefficients allow a more precise mapping between specific design decisions and their motivational outcomes, as illustrated in Fig. 1 and summarized in Table 1. The finding that learning anxiety more strongly predicted I-type curiosity ($\beta = .62$) than D-type curiosity ($\beta = .54$) carries an important design implication the original manuscript did not sufficiently address. This pattern suggests that when anxiety-inducing contexts are paired with appropriate technological scaffolding, specifically the autonomous navigation mode and sensor-driven real-time feedback afforded by the AR application, learners do not merely seek to close knowledge gaps in a tension-driven manner, but more readily transition into a state of pleasurable, intrinsically motivated exploration. The design consequence is that information gap creation alone is insufficient; it must be coupled with an environment that communicates competence-affirming feedback and preserves learner agency over exploration paths. In the present application, this was achieved through the combination of incomplete mission cards (inducing the gap) and color-coded AR overlays responsive to IoT sensor readings (validating discoveries and sustaining autonomy), a pairing the empirical data confirm to be more effective at facilitating I-type curiosity than at merely sustaining D-type tension.

The differential contribution of the two curiosity pathways to flow experience, with I-type showing a slightly stronger effect ($\beta = .44$) than D-type ($\beta = .38$) and both together explaining 52% of the variance in flow, further clarifies a design priority that existing literature has not adequately specified. While both curiosity types contribute to flow induction, the relatively stronger I-type pathway suggests that design efforts should prioritize features sustaining pleasurable exploration over those merely maintaining tension-driven information seeking. In the application, this priority is reflected in the real-time animated achievement feedback and progressive task disclosure mechanisms, which convert the act of discovery into an intrinsically rewarding experience rather than compliance-driven task completion. The moderate observed flow levels ($M = 2.83$) confirm that even brief, theoretically grounded AR exposure is sufficient to activate the curiosity–flow transition, while also indicating that extended implementations allowing greater skill development would likely yield more robust flow states, providing actionable guidance for practitioners adapting the framework to longer curricula. (see Table 5)

5.6. Limitations and future directions

Several limitations warrant consideration. The single-group design limits causal inferences about the curriculum's relative effectiveness compared to traditional instruction. While our theoretical approach

Table 5
Theory-design-evidence alignment matrix.

Motivational Stage	Theoretical Mechanism	AR-IoT Design Feature	Empirical Finding
Anxiety Induction	Deliberate information gap creation	Incomplete mission cards; partial habitat cues withheld	Anxiety→ D-type $\beta = .54$; Anxiety→ I-type $\beta = .62$
Curiosity Transformation	Autonomy + real-time sensor feedback	Free-navigation mode; IoT-driven color-coded AR overlays	D-type→ Flow $\beta = .38$; I-type→ Flow $\beta = .44$; $R^2 = .52$
Flow Facilitation	Challenge-skill balance + achievement visualization	Difficulty-tiered task sequence; animated achievement screen	Flow→ Continuance Intention $\beta = .53$; $R^2 = .28$

Note. β values are standardized path coefficients from the structural equation model. R^2 values indicate variance explained in the respective endogenous construct.

successfully predicted psychological mechanisms, controlled comparisons are necessary to establish superior learning outcomes. The sample comprised fifth-graders from one region, limiting generalizability across ages and cultural contexts.

Reliance on self-report measures introduces potential biases. Future research should incorporate behavioral data (exploration patterns, time-on-task), physiological indicators (eye-tracking, electrodermal activity), and learning outcome assessments to triangulate psychological state measurements. The moderate flow levels suggest that longer interventions allowing skill development may yield stronger effects.

Finally, translating these findings to sustainable educational programs requires addressing practical constraints including teacher training needs, technology infrastructure requirements, and curriculum integration challenges. Implementation research examining how schools adapt and scale such interventions would provide essential insights for widespread adoption.

A further consideration concerns the replicability of the AR-IoT system itself. While the theoretical framework and measurement instruments employed in this study are fully replicable using standard AR development platforms and validated psychometric scales, the physical IoT infrastructure requires institutional resources, including campus Wi-Fi coverage, sensor installation permissions, and device provisioning, that may not be uniformly available across all educational settings. Researchers seeking to replicate the motivational mechanisms without full IoT infrastructure may adopt a simplified approach by substituting live sensor feeds with pre-recorded environmental data overlays, which preserves the core AR interaction features while reducing hardware dependency. Alternatively, the anxiety-curiosity-flow sequence can be operationalized through other AR platforms (e.g., AR Core-based applications) using equivalent information-gap and free-navigation design principles, as these constitute the theoretically active mechanisms rather than the specific technology stack. The complete trigger marker image set, sensor configuration files, and Unity project scene files are archived and available upon reasonable request to support adaptation across diverse institutional contexts.

6. Conclusion

Augmented reality, when grounded in integrated motivational theory, demonstrates strong potential as a pedagogical tool for environmental education. By intentionally designing for the transition from anxiety to curiosity and from curiosity to flow, the curriculum activated psychological processes that sustain learner engagement with ecological issues. This validated framework provides a replicable reference for creating technology-enhanced learning experiences that move beyond information delivery toward intrinsically motivated discovery.

The results highlight that effective educational technology requires deliberate orchestration of digital affordances to support theoretically informed psychological mechanisms. The success of the hybrid AR approach, combining physical manipulation with digital augmentation, demonstrates the value of design strategies that respect embodied learning while enhancing it through computational capabilities. More broadly, the study shows that purposefully activating complementary motivational mechanisms through digital design can foster learning

experiences that learners not only benefit from but are eager to sustain. Such outcomes provide a critical foundation for addressing the long-term challenges of environmental education, where enduring motivation is as vital as knowledge acquisition.

CRedit authorship contribution statement

ChuYu Cheng: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **ChiaHui Feng:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Chu Yu Cheng reports financial support was provided by National Science and Technology Council, R.O.C. Reports a relationship with that includes:. Has patent pending to. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] UNESCO. The salamanca statement and framework for action on special needs education. Paris: United Nations Educational, Scientific and Cultural Organization; 1994. editor.

[2] Litman J. Curiosity and the pleasures of learning: wanting and liking new information. *Cogn Emot* 2005;19(6):793–814. <https://doi.org/10.1080/02699930541000101>.

[3] Litman JA. Interest and deprivation factors of epistemic curiosity. *Pers Individ Dif* 2008;44(7):1585–95. <https://doi.org/10.1016/j.paid.2008.01.014>.

[4] Csikszentmihalyi M. Finding flow: the psychology of engagement with everyday life. 1st Edition editor. Basic Books; 1998.

[5] Santos WOD, Bittencourt II, Isotani S, Dermeval D, Marques LB, Silveira IF. Flow theory to promote learning in educational systems: is it really relevant? *Rev Bras Inform na Educ* 2018;26:29.

[6] Alan S, Mumcu I. Nurturing childhood curiosity to enhance learning: evidence from a randomized pedagogical intervention. *Am Econ Rev* 2024;114(4):1173–210. <https://doi.org/10.1257/aer.20230084>.

[7] Murayama K, FitzGibbon L, Sakaki M. Process account of curiosity and interest: a reward-learning perspective. *Educ Psychol Rev* 2019;31(4):875–95. <https://doi.org/10.1007/s10648-019-09499-9>.

[8] Oudeyer PY, Gottlieb J, Lopes M. Chapter 11 - intrinsic motivation, curiosity, and learning: theory and applications in educational technologies. In: Studer B, Knecht S, editors. *Progress in brain research. Progress in brain research*, 229. Elsevier; 2016. p. 257–84.

[9] Csikszentmihalyi M. Flow: the psychology of optimal experience. Harper & Row; 2009.

[10] Gupta S, Priyanka. Gamification and e-learning adoption: a sequential mediation analysis of flow and engagement. *VINE J Inf Knowl Manag Syst* 2024;54(6): 1342–59. <https://doi.org/10.1108/vjkm-04-2022-0131>.

[11] Özhan ŞÇ, Kocadere SA. The effects of flow, emotional engagement, and motivation on success in a gamified online learning environment. *J Educ Comput Res* 2019;57:2006–31.

[12] Esteban-Millat I, Martínez-López FJ, Huertas-García R, Meseguer-Artola A, Rodríguez-Ardura I. Modelling students' flow experiences in an online learning environment. *Comput Educ* 2014;71:111–23.

[13] Chen Y, Wang Q, Chen H, Song X, Tang H, Tian M. An overview of augmented reality technology. *J Phys: Conf Ser* 2019;1237(2):022082. <https://doi.org/10.1088/1742-6596/1237/2/022082>.

- [14] Fonseca X, Spangenberg P, Baer M, Schmidt R, Söbke H. Location-based augmented reality in education: a systematic literature review. *Comput Educ Open* 2025;9:100277. <https://doi.org/10.1016/j.caeo.2025.100277>.
- [15] Choi DH, Kim J, Kim SH. ERP training with a web-based electronic learning system: the flow theory perspective. *Int J Hum Comput Stud* 2007;65(3):223–43. <https://doi.org/10.1016/j.ijhcs.2006.10.002>.
- [16] Goh T-T, Yang B. The role of e-engagement and flow on the continuance with a learning management system in a blended learning environment. *Int J Educ Technol High Educ* 2021;18(1):49. <https://doi.org/10.1186/s41239-021-00285-8>.
- [17] Huang B, Hew KF, Lo CK. Investigating the effects of gamification-enhanced flipped learning on undergraduate students' behavioral and cognitive engagement. *Interact Learn Environ* 2019;27(8):1106–26. <https://doi.org/10.1080/10494820.2018.1495653>.
- [18] Einstein I. Curiosity. In: Linnenbrink-Garcia RPaL, editor. *International handbook of emotions in education*. New York: Routledge; 2014. p. 228–45.
- [19] Pekrun R. The control-value theory of achievement emotions: assumptions, corollaries, and implications for educational research and practice. *Educ Psychol Rev* 2006;18(4):315–41. <https://doi.org/10.1007/s10648-006-9029-9>.
- [20] Loewenstein G. The psychology of curiosity: a review and reinterpretation. *Psychol Bull* 1994;116(1):75–98. <https://doi.org/10.1037/0033-2909.116.1.75>.
- [21] Matthias AV, Gruber J, Ranganath Charan. *Curiosity and learning: a neuroscientific perspective*. The cambridge handbook on motivation and learning. Cambridge University Press; 2019.
- [22] Lamnina M, Chase CC. Developing a thirst for knowledge: how uncertainty in the classroom influences curiosity, affect, learning, and transfer. *Contemp Educ Psychol* 2019;59:101785. <https://doi.org/10.1016/j.cedpsych.2019.101785>.
- [23] Schutte NS, Malouff JM. Connections between curiosity, flow and creativity. *Pers Individ Dif* 2020;152:109555. <https://doi.org/10.1016/j.paid.2019.109555>.
- [24] Abdelghani R, Oudeyer P-Y, Law E, de Vulpillières C, Sauzéon H. Conversational agents for fostering curiosity-driven learning in children. *Int J Hum Comput Stud* 2022;167:102887. <https://doi.org/10.1016/j.ijhcs.2022.102887>.
- [25] Litman JA, Jimerson TL. The measurement of curiosity as a feeling of deprivation. *J Pers Assess* 2004;82(2):147–57. https://doi.org/10.1207/s15327752jpa8202_3.
- [26] Stern M, Powell R, Ardoin N. What difference does it make? Assessing outcomes from participation in a residential environmental education program. *J Env Educ* 2008;39:31–43. <https://doi.org/10.3200/JOEE.39.4.31-43>.
- [27] Hu Lt, Bentler PM. Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. *Struct Equ Model: Multidiscip J* 1999;6(1):1–55. <https://doi.org/10.1080/10705519909540118>.
- [28] Byrne BM. *Structural equation modeling with AMOS: basic concepts, applications, and programming*. 3rd Edition editor. Routledge; 2016.
- [29] Khan T, Johnston K, Ophoff J. The impact of an augmented reality application on learning motivation of students. *Adv Hum-Comput Interact* 2019;2019(1):1–14. <https://doi.org/10.1155/2019/7208494>.
- [30] Hong J-C, Hwang M-Y, Tai K-H, Lin P-H. Improving cognitive certitude with calibration mediated by cognitive anxiety, online learning self-efficacy and interest in learning chinese pronunciation. *Educ Technol Res Dev* 2019;67(3):597–615. <https://doi.org/10.1007/s11423-018-9628-4>.